Answer key: Lesson 9, The Problem-Solving Toolbox

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Use a strip. Kumar has 12 marbles. Ravi has 5 fewer marbles than Kumar.

Answer: 7 marbles; 19 marbles in all

Ravi: 12 take away 5 is 7 marbles. In all: 12 + 7 = 19 marbles.

Watch for: Writing 17 for Ravi (12 and 5 added).; Writing 12 + 5 for the total.

S2. Do one step at a time. A shop has 30 apples. It sells 12 apples in the morning and 9 apples in the afternoon.

Answer: 18 apples; 9 apples

After the morning: 30 take away 12 is 18 apples. After the afternoon: 18 take away 9 is 9 apples.

Watch for: Taking 12 and 9 away from 30 in one go and writing 9 for both parts.; Adding 12 and 9.

S3. Make a table. Each row has 3 more beads than the row before. Fill in the table.

Answer: 15 beads; 18 beads

3, 6, 9, 12, 15, 18. Row 5 has 15 beads and row 6 has 18 beads.

Watch for: Writing 12 for row 5 (stopping one box early).; Writing 5 times 3 for row 6.

S4. Work backwards. Ravi thinks of a number. He takes away 8 and gets 17.

Answer: 25; 17

17 and 8 make 25, so Ravi's number is 25. Check: 25 take away 8 is 17.

Watch for: Writing 9 (17 take away 8).; Not checking.

S5. Find one first. 4 bags hold 20 apples. All the bags hold the same number of apples.

Answer: 5 apples; 35 apples

20 shared into 4 bags is 5 apples in 1 bag. 7 bags: 7 groups of 5 is 35 apples.

Watch for: Writing 24 (20 + 4).; Writing 7 + 5 for 7 bags.

S6. Use a list. Ali has three coins: 5¢, 10¢ and 20¢. He picks 2 coins. Fill in the blanks.

Answer: 15¢, 25¢, 30¢; 3 pairs

5¢ and 10¢ make 15¢. 5¢ and 20¢ make 25¢. 10¢ and 20¢ make 30¢. There are 3 different pairs.

Watch for: Adding all three coins.; Writing 6 pairs (counting 5¢ and 10¢, then 10¢ and 5¢ again).

Think harder (page 3)

T1. Read the story. 3 rulers cost $15. All the rulers cost the same. Siti buys 5 rulers.

Answer: Find one first; $25

Find one first: $15 shared into 3 rulers is $5 for 1 ruler. 5 rulers cost 5 groups of $5, which is $25.

Watch for: Choosing Table (the number of rulers asked, 5, is not in the pattern 3, 6, 9).; Writing $20 (15 + 5).

T2. Ben pays with a $10 note for a toy and a card. The toy costs $6. His change is $1.

Answer: $9; $3

Ben spent $10 take away $1, which is $9. The toy cost $6, so the card costs $9 take away $6, which is $3. Check: 6 + 3 = 9 and 10 take away 9 is 1.

Watch for: Writing $4 (10 take away 6).; Writing $5 (6 and 1 taken from 10 the wrong way).

T3. Siti is 4th from the front of a line. She is also 3rd from the back. How many children are in the line? Use the box to show your thinking.

Answer: 3 children; 6 children

There are 3 children in front of Siti. There are 2 children behind Siti, because she is 3rd from the back. 3 + 1 + 2 = 6 children. Siti is counted once.

Watch for: Writing 7 (4 + 3, counting Siti twice).; Writing 5.

T4. Mei has 7 coins. Some are 5¢ coins and some are 10¢ coins. The coins make 50¢ in all. You may use the table.

Answer: 4 coins of 5¢; 3 coins of 10¢

Try 2 coins of 10¢: 5 coins of 5¢ make 25, and 20 + 25 is 45, too little. Try 3 coins of 10¢: 4 coins of 5¢ make 20, and 30 + 20 is 50. So 4 coins of 5¢ and 3 coins of 10¢.

Watch for: Guessing without checking the total.; Writing 3 and 4 the wrong way round.

Go further (page 4)

G1. Jenny has 3 times as many stickers as Jason. They have 36 stickers in all. Solve it two ways: a strip and a table of tries.

Answer: 9 stickers; 27 stickers; Yes

Road 1, the strip: Jason has 1 part and Jenny has 3 parts, so there are 4 equal parts. 36 shared into 4 is 9, so Jason has 9 stickers. Road 2, the table: Jason 8 gives 32 in all, and Jason 10 gives 40 in all. Try Jason 9: Jenny has 27 and 9 + 27 = 36. Both roads say Jason has 9. Jenny has 27.

Watch for: Writing 12 for Jason (36 shared into 3 parts).; Trying numbers in a different order and losing track.

G2. Ali has 9 stickers, Ben has 15 stickers and Mei has 12 stickers. Only Ben and Ali move stickers. Then all three children have the same number.

Answer: 12 stickers; 3 stickers

9 + 15 + 12 = 36 stickers. 36 shared into 3 children is 12 each. Mei already has 12. Ben has 3 too many and Ali has 3 too few, so Ben gives 3 stickers to Ali.

Watch for: Making Ali and Ben equal without looking at Mei.; Writing 6 (the difference between 9 and 15).

G3. Siti reads 4 pages on day 1, 7 pages on day 2 and 10 pages on day 3. Each day she reads 3 more pages than the day before. Finish the sentence about the tool you used.

Answer: Day 7; many answers are right

A table: 4, 7, 10, 13, 16, 19, 22. She reads 22 pages on day 7. Or work backwards: 22 take away 4 is 18, and 18 is 6 steps of 3, so day 7. Check: 6 steps from day 1 is day 7.

Watch for: Writing 6 (the number of steps, not the day).; Not saying why.

Investigation (page 5): Pick a tool

Many answers are right. A good problem has a story, numbers, one question, and an answer that fits the tool. The check uses a second tool or works the problem forwards.

Look for: A child who writes a short story with numbers and a question, solves it with the chosen tool, and checks it a second way (a different tool, or working forwards). Credit any clear problem. A child who copies a card problem has not yet made their own: ask, what would you change? A child who uses a number that gives a whole answer has thought about the check.

Exit ticket (page 5)

E1. Mei thinks of a number. She adds 4 and gets 15.

Answer: Work backwards

We know the end (15) and not the start, so we work backwards: 15 take away 4 is 11.

Watch for: Choosing Table (no pattern grows).; Choosing List (we do not list every way).

E2. Ali has 11 stickers. Mei has 4 fewer stickers than Ali.

Answer: 7 stickers; 18 stickers

Mei: 11 take away 4 is 7. In all: 11 + 7 = 18.

Watch for: Writing 15 (11 and 4 added).; Leaving out one strip when finding the total.

E3. 5 buns cost $20. A child in another class says 1 bun costs $5.

Answer: False

$20 shared into 5 buns is $4 for 1 bun, not $5. Check: 5 buns at $5 would cost $25.

Watch for: Choosing True (using the 5 from 5 buns).; Dividing 20 by 4.

Sneak-peek puzzles (not in the workbook)

1. 7 stickers.
2. 4 stickers. Mei 4 and Ali 6 make 10, and 6 is 2 more than 4.